

Working

1. PIR sensor detects human or vehicle movement.
 2. Arduino receives the sensor signal.
 3. When motion is detected:
 - All street-light LEDs turn ON.
 4. When motion stops:
 - LEDs turn OFF after the next sensor reading.
 5. This saves electricity compared to keeping lights ON continuously.
-

Advantages

- Low cost
- Easy implementation
- Energy efficient
- Automatic operation
- Suitable for smart city applications